

JINDAL STAINLESS

ENGINEERING CASE STUDY COMPETITION 2026

PROBLEM STATEMENTS & GUIDELINES

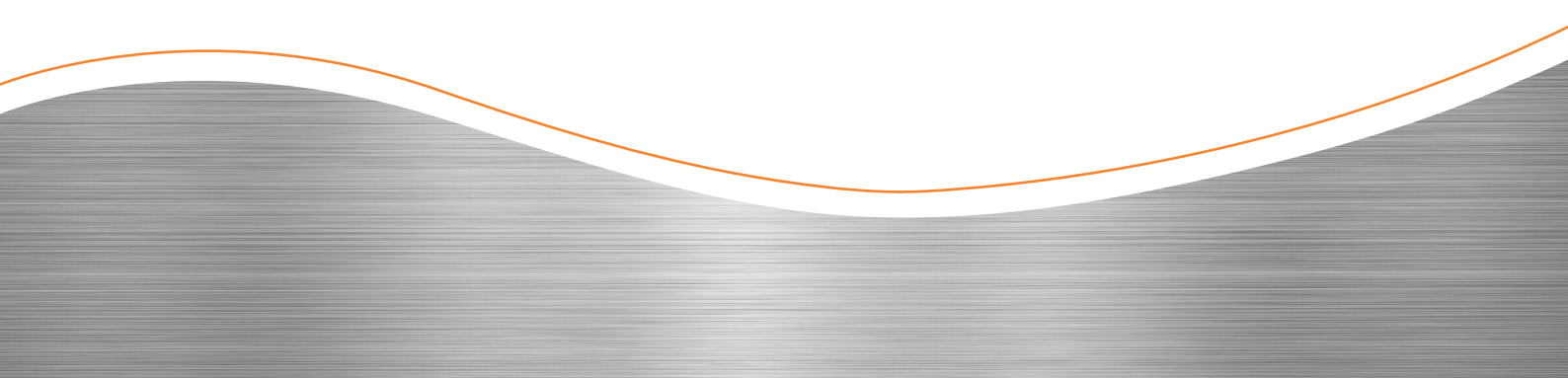
Innovation • Sustainability • Digital Manufacturing • Engineering Excellence



About Jindal Stainless

Jindal Stainless is India's largest stainless steel manufacturer and one of the world's leading stainless steel producers, with an annual melt capacity of over 3 million tonnes. The Company manufactures a comprehensive portfolio of stainless steel flat and long products catering to infrastructure, railways, automotive, architecture, process industries, defence, renewable energy, consumer goods and precision engineering.

With integrated manufacturing facilities at Jajpur and Hisar, Jindal Stainless combines advanced manufacturing, innovation and sustainability to deliver world-class products across global markets while creating value for customers through quality, technology and responsible manufacturing.



Problem Statement 1

AI-Powered Surface Defect Detection

Business Context

Surface quality is one of the most critical quality parameters in stainless steel manufacturing. Jindal Stainless continuously focuses on improving product quality, manufacturing efficiency and customer satisfaction through digital technologies and smart manufacturing initiatives. Intelligent inspection systems have the potential to significantly improve quality assurance while reducing waste and operational losses.

Background

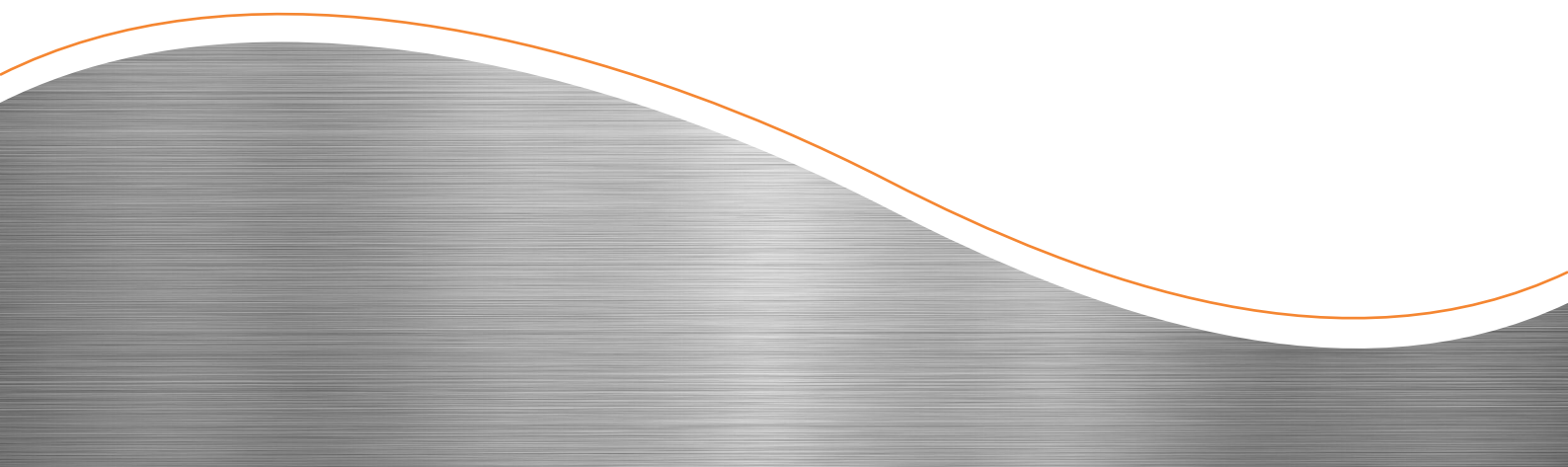
Surface and edge defects such as scratches, scale, roll marks and edge cracks are often detected late on the rolling and finishing lines. Late detection leads to yield loss, downgrading, rework and customer rejections, while manual inspection is slow, subjective and hard to scale at high line speeds.

Action Item

Develop a computer-vision model that detects and classifies surface defects on steel-strip images in real time. Teams may use publicly available steel surface-defect datasets to train and validate their model, and should deliver a working demo that flags and labels defects on an uploaded image, ideally with a confidence score and defect type.

Key Considerations

- Detection accuracy and false-alarm rate
- Ability to handle multiple defect types
- Inference speed suitable for a production line
- A usable demo interface
- A clear path to scaling across grades and products



Problem Statement 2

Stainless Steel Grade Selector

Business Context

Jindal Stainless manufactures a wide range of stainless steel grades serving applications across infrastructure, architecture, mobility, railways, process industries, renewable energy and consumer products. Selecting the appropriate grade is essential for ensuring performance, durability and lifecycle value while optimizing costs for customers.

Background

Choosing the right stainless steel grade for a given application depends on factors such as corrosion environment, operating temperature, strength requirements, formability and cost. Engineers, fabricators and new customers often lack an easy, guided way to make this choice, which can lead to over-specification or poor material fit.

Action Item

Build a recommender tool (web or mobile app) where a user enters their application requirements and receives the best-fit stainless steel grade or grades, along with the trade-offs (for example corrosion resistance versus cost). The tool should explain why each grade is suggested and highlight key properties.

Key Considerations

- Accuracy and sensibility of recommendations
- Quality of the underlying grade-property logic or data
- Clarity of the trade-off explanation
- Ease of use
- Potential to extend to new applications and markets

Problem Statement 3

Carbon and Energy Calculator for Steelmaking

Business Context

Sustainability and responsible manufacturing are integral to Jindal Stainless' long-term growth strategy. Increasing transparency around carbon emissions and energy consumption can help accelerate informed decision-making and support the industry's transition towards lower-carbon steelmaking.

Background

Stainless steel making is energy- and emissions-intensive, and carbon intensity per tonne depends heavily on the input mix and energy sources used. With a Net Zero by 2050 commitment, there is a need for simple tools that make the emissions impact of different choices visible and help identify lower-carbon options.

Action Item

Build a tool that estimates carbon emissions per tonne of steel based on a chosen input mix (for example share of scrap versus virgin input) and energy source and let the user adjust these inputs to see the effect on emissions. The tool should help users find combinations that minimise CO₂ per tonne while remaining practical.

Key Considerations

- Reasonableness of the emissions model and assumptions
- Clarity of how inputs affect the result
- An interactive, easy-to-understand interface
- The usefulness of the optimisation or comparison feature
- A measurable reduction in estimated emissions

Problem Statement 4

Material Selection Solution for Solar Mounting Structures

Business Context

The rapid growth of renewable energy infrastructure presents opportunities for advanced materials capable of delivering long-term durability in harsh service environments. Stainless steel has emerged as an important engineering material due to its corrosion resistance, structural performance and lifecycle benefits.

Background

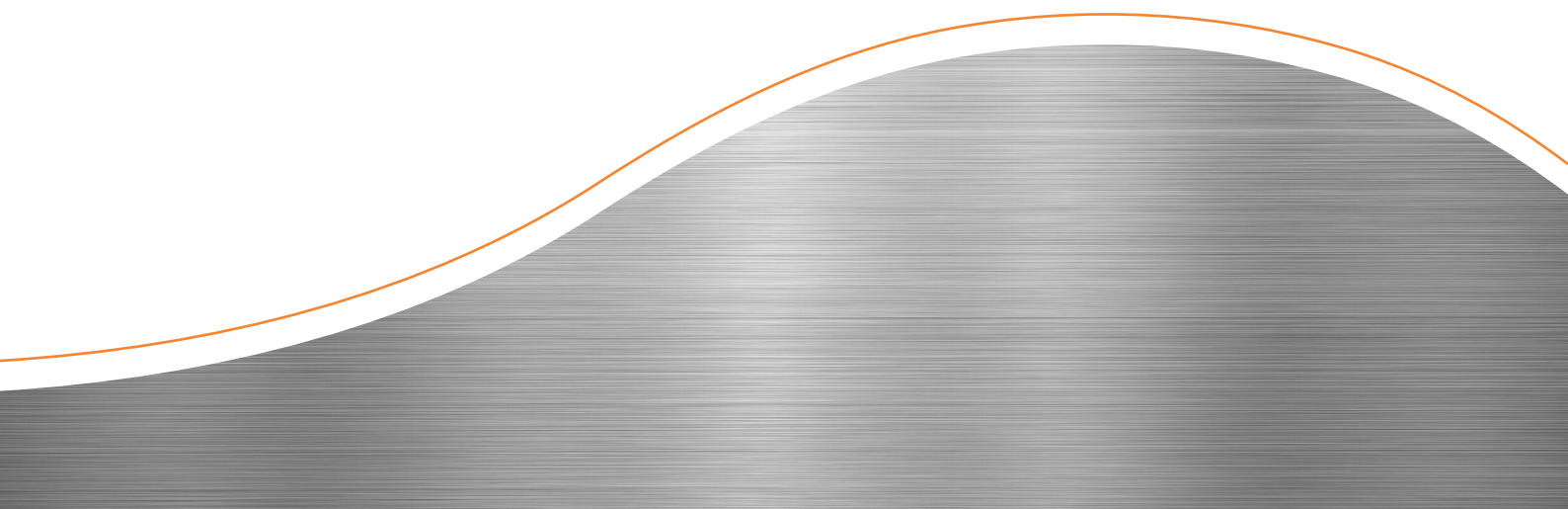
Develop a technically robust and cost-effective material selection solution for solar panel mounting structures intended for deployment in highly corrosive coastal or saline environments. The structures will be exposed to severe service conditions, including high chloride levels in soil and atmosphere, seasonal water-logging during the monsoon, extreme dryness during summer and significant wind loads.

The study should evaluate stainless steel, galvanized steel, pre-painted galvanized steel, magnesium-zinc-aluminium coated steel, FRP and any other suitable alternative materials for corrosion resistance, structural integrity, durability, lifecycle cost, and maintenance requirements.

The objective is to recommend the most suitable material and design approach that can ensure a reliable service life of at least 25–30 years with minimal maintenance, while remaining commercially viable for solar panel manufacturers and contractors.

Key Considerations

- Corrosion resistance
- Structural integrity
- Lifecycle cost
- Commercial viability
- Long-term sustainability



Case Guidelines

Each participating team should submit a presentation comprising 4–5 slides that clearly communicates the proposed solution. The deck should be structured to demonstrate technical depth, business relevance and implementation feasibility.

Problem Understanding: Define the problem, explain its significance and establish the need for the solution.

Objectives: State the expected outcomes and measurable success indicators.

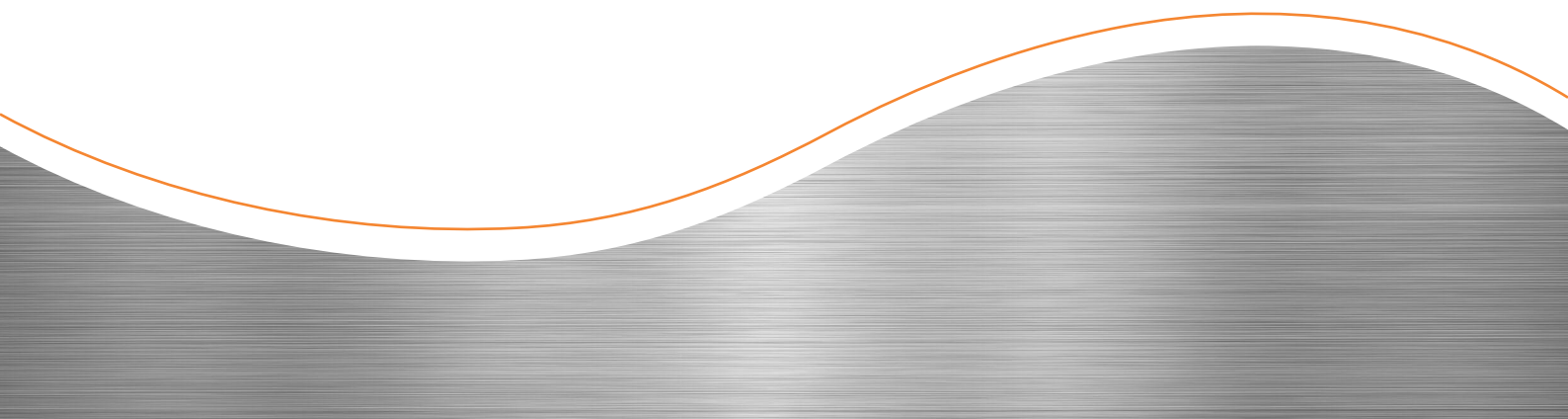
Proposed Solution: Describe the solution, research done, methodology, architecture, workflow, implementation plan and expected outcomes.

Validation & Feasibility: Present supporting evidence through calculations, simulations, datasets, prototypes or proof of concept.

Assumptions & Limitations: Highlight assumptions and implementation constraints.

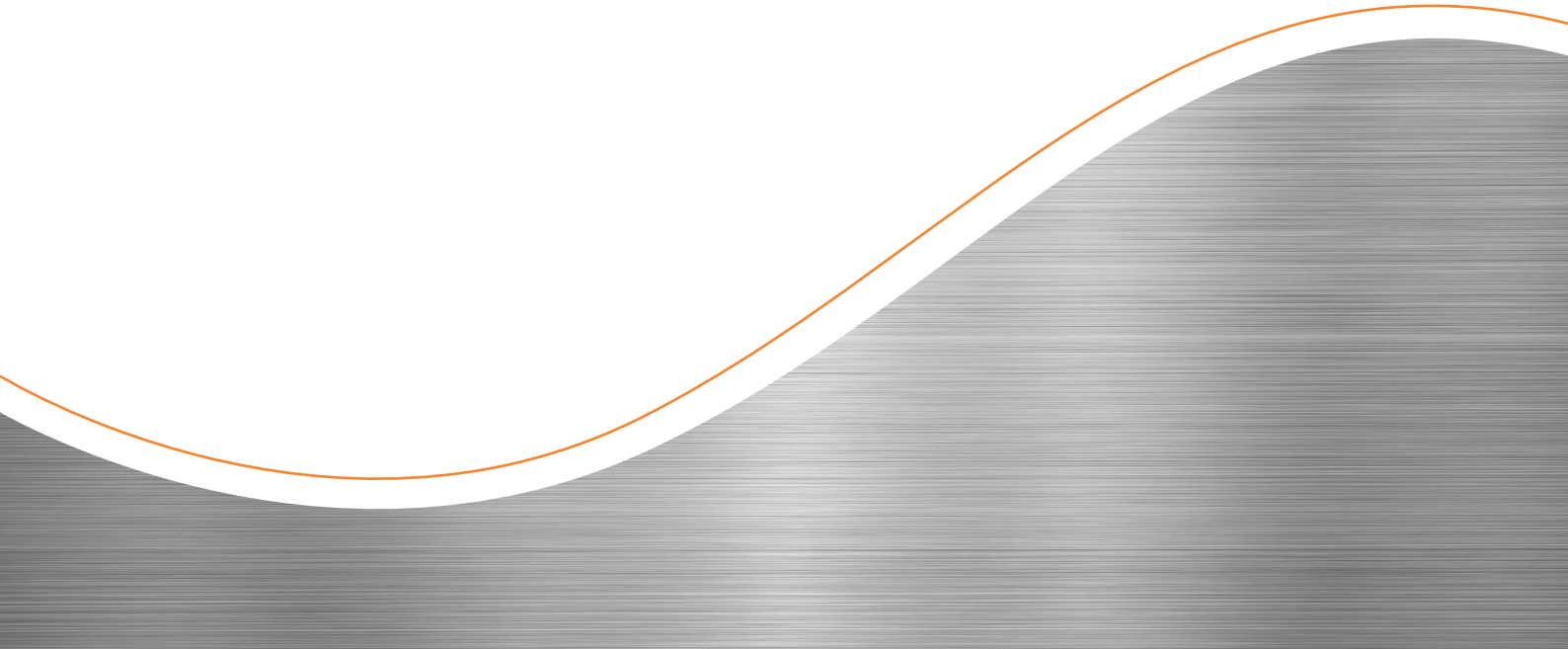
Innovation & Differentiation: Explain how the solution improves upon existing approaches.

Expected Impact: Summarise the technical, operational, commercial and sustainability benefits.



Evaluation Criteria

Evaluation Parameter	Assessment Focus
Problem Understanding	Depth of understanding of the challenge.
Innovation & Originality	Novelty and creativity of the idea.
Technical Excellence	Engineering robustness and feasibility.
Business Relevance & Impact	Potential value to Jindal Stainless and customers.
Feasibility & Scalability	Industrial applicability and scalability.
Presentation & Communication	Clarity, structure and effectiveness.



THANK YOU

